THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

OBJECT ORIENTED PROGRAMMING

ASSIGNMENT NO.2

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2O31

1. Create a structure in C++ containing the details of Students as details below and a main function to execute the structure.

Data Members(properties):

Name

Roll No

Degree

Hostel

CurrentCGPA

Member Function(behavior):

addDetails(); updateDetails();

updateCGPA(); updateHostel();

displaydetails();

CODE:

```
#include <iostream>
#include <string>
using namespace std;
struct Student
{
   string name;
  int roll_no;
  float cgpa;
 string degree;
 string hostel;
    void addDetails()
```

```
    {
        cout << "Enter name: ";
        cin >> name;

        cout << "Enter roll number: ";
cin >> roll_no;

        cout << "Enter CGPA: ";
cin >> cgpa;

        cout << "Enter degree: ";
cin >> degree;

        cout << "Enter hostel: ";
cin >> hostel;
    }

    void updateDetails()
    {
        cout << "\nEnter new name: ";
cin >> name;

        cout << "Enter new roll number: ";
cin >> roll_no;

        cout << "Enter new CGPA: ";
cin >> cgpa;
```

Output

```
Enter name: Ridhika
Enter roll number: 280
Enter CGPA: 7.6
Enter degree: BTech
Enter hostel: N

Student Details:
Name: Ridhika
Roll No: 280
CGPA: 7.6
Degree: BTech
Hostel: N

Enter new name: Jaibhev
Enter new roll number: 281
Enter new CGPA: 9
Enter new degree: BTech
Enter new hostel: M


=== Code Execution Successful ===
```

```
        cout << "Enter new degree: ";
cin >> degree;

        cout << "Enter new hostel: ";
cin >> hostel;
    }




    void displayDetails()
    {
        cout << "\nStudent Details:\n";
cout << "Name: " << name << endl;
cout << "Roll No: " << roll_no << endl;
cout << "CGPA: " << cgpa << endl;        cout
<< "Degree: " << degree << endl;        cout
<< "Hostel: " << hostel << endl;
    }
}s;

int main()
{

    s.addDetails();
    s.displayDetails();
    s.updateDetails();
return 0;
}
```

2. Differentiate between private and public access/scope. Perform the question no. 1 with class instead of structure with having the data members private and some member functions in private scope and some in public scope

CODE:

```
#include <iostream>

#include <string> using

namespace std;


class Student

{

private:

   string name;

int roll_no;     float

cgpa;     string

degree;     string

hostel;


public:     void addDetails()

   {

      cout << "Enter name: ";        cin

>> name;
```

```
        cout << "Enter roll number: ";
cin >> roll_no;

        cout << "Enter CGPA: ";        cin
>> cgpa;

        cout << "Enter degree: ";        cin
>> degree;

        cout << "Enter hostel: ";
cin >> hostel;
    }

    void updateDetails()
    {
        cout << "Enter new name: ";
cin >> name;

        cout << "Enter new roll number: ";
cin >> roll_no;

        cout << "Enter new CGPA: ";
cin >> cgpa;

        cout << "Enter new degree: ";
cin >> degree;
```

```
        cout << "Enter new hostel: ";
cin >> hostel;
    }

    void displayDetails()
    {
        cout << "\nStudent Details:\n";
cout << "Name: " << name << endl;
cout << "Roll No: " << roll_no << endl;
cout << "CGPA: " << cgpa << endl;        cout
<< "Degree: " << degree << endl;        cout
<< "Hostel: " << hostel << endl;
    }
};
int main()
{
    Student s;
    s.addDetails();
    s.displayDetails();
    s.updateDetails();
    s.displayDetails();
    return 0;
}
```

Output

```
Enter name: Ridhika
Enter roll number: 280
Enter CGPA: 7.6
Enter degree: BTech
Enter hostel: N

Student Details:
Name: Ridhika
Roll No: 280
CGPA: 7.6
Degree: BTech
Hostel: N

Enter new name: Jaibhev
Enter new roll number: 281
Enter new CGPA: 9
Enter new degree: BTech
Enter new hostel: M

Student Details:
Name: Jaibhev
Roll No: 281
CGPA: 9
Degree: BTech
Hostel: M


=== Code Execution Successful ===
```

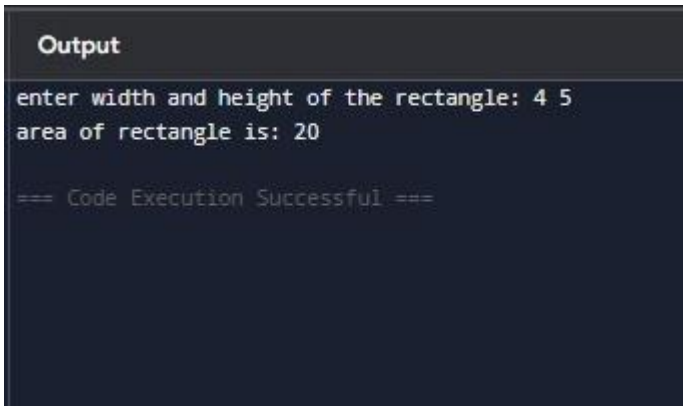
3. Define a class Rectangle with variables width and height of integer typwe along with following:

(a) void getdata() to initialize object values

(b) void calculatearea() to calculate and display the area.

CODE

```
#include<iostream>
using namespace std;
class rectangle{
private:
int width,height;
public:
 void getdata(){
        cout<<"enter width and height of the rectangle: ";
        cin>>width>>height;
}
        void calculatearea(){
        int area=width*height;
cout<<"area of rectangle is: "<<area;
        }
};
int main(){
rectangle rect;
 rect.getdata();
rect.calculatearea();
return 0;
}
```



4. Define a class Complex with variables real and imaginary along with following:

(a)void setComplex (float, float) to initialize object values.

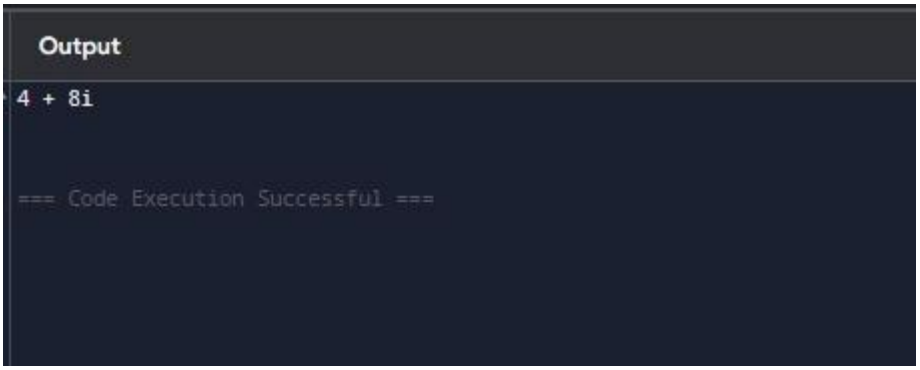
(b) void displayComplex() to show the complex number

(c)Pass and return objects to calculate sum of two complex numbers. Display the sum

CODE:

```
#include <iostream>
using namespace std;

class Complex {
float real, imag;

public:     void setComplex(float
r, float i)
    {        real
= r;
imag = i;
    }

    void displayComplex()
    {
        cout << real << " + " << imag << "i" << endl;
    }

    Complex add(Complex c)
```

```
    {
        Complex temp;        temp.real
= real + c.real;        temp.imag =
imag + c.imag;        return temp;
    }
};

int main()
{
    Complex c1, c2, sum;

    c1.setComplex(2.5, 3.5);
c2.setComplex(1.5, 4.5);

    sum = c1.add(c2);

    sum.displayComplex();

    return 0;
}
```



5. Implement scope resolution operator : : for the following uses:

(a)Class functions defined outside the class

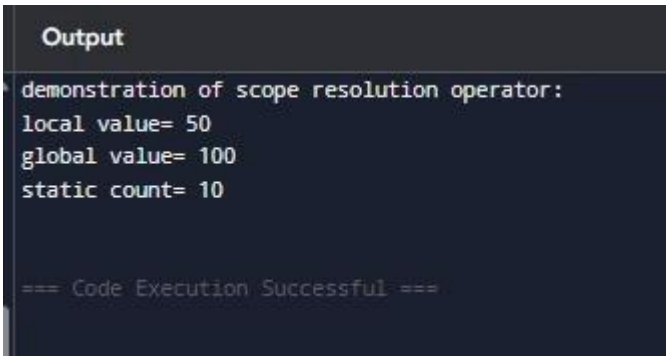
(b) Access a global variable with same name as a local variable

(c)Access a static variables

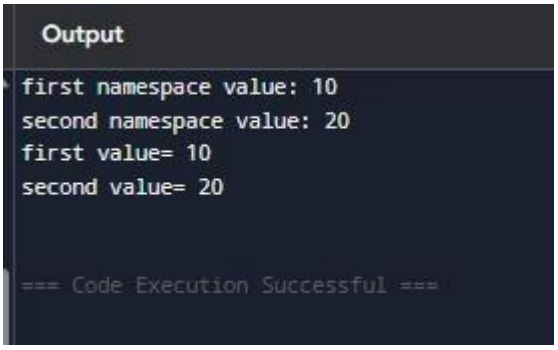
(d) Use inbuilt libraries (cin cout with scope resolution operator)

<u>-CODE:</u>

```
#include <iostream> using
namespace std;

int value = 100;

class demo
{ private:
static int
count;

public:
   void show();
};

void demo::show()
{     int value = 50;     cout << "local value=
" << value << endl;     cout << "global value= " <<
::value << endl;     cout << "static count= " <<
demo::count << endl;
}

int demo::count = 10;

int main() {     demo d;     cout << "demonstration of scope
resolution operator:" << endl;     d.show();     return 0;
}
```



6. Create a code to implement the namespace and use similar variables and functions defined in different code sections.

CODE:

```
#include <iostream> using namespace std;


namespace first {     int value = 10;     void show()
    {
        cout << "first namespace value: " << value <<
endl;
    }
}


namespace second
{     int value =
20;     void
show()
    {
        cout << "second namespace value: " << value << endl;
    }
}


int main() {     first::show();     second::show();     cout
<< "first value= " << first::value << endl;     cout <<
"second value= " << second::value << endl;     return
0;
}
```